DATA ITEM DESCRIPTION

Title: AIRFRAME RIGIDITY TEST REPORT

Number: DI-MISC-81387A
Approved Date: 20070927
AMSC Number: F9003
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Office of Primary Responsibility: 11 (ASC/ENFS)
Applicable Forms: N/A

Use/Relationship: This report on Airframe Rigidity Testing will provide data for the detailed evaluation of the air vehicle modal vibration characteristics and rigidity properties and comparison with calculated results. The principal use is to validate or update analytical models.

a. This Data Item Description (DID) is applicable to any contract for which a ground vibration test or other airframe rigidity test is required.

b. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in the contract.

c. DI-MISC-81387A supersedes DI-MISC-81387.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices and revisions shall be as specified in the contract.

2. Format. The Airframe Rigidity Test Report shall be in the contractor's format.

3. Content. The Airframe Rigidity Test Report shall contain:

3.1. The Air vehicle Vibration Test. When applicable, this section shall contain the following:

a. Descriptions and photographs of the air vehicle suspension system, the instrumentation, and the air vehicle configurations.

b. Description of test procedures, data acquisition, and methods of data reduction analysis.

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c. Shaker locations.

d. Mode shape measurement locations.

e. Plots of the transfer function versus frequency for strategically located vibration pickups.

f. The natural frequency, mode shapes, structural damping, and node line locations for all obtained structural modes of vibration that may be important with respect to flutter and other dynamic analyses.

g. A comparison of the measured air vehicle modal data (frequencies, mode shapes, and node lines) with those as predicted by the analytical model.

3.1 Aeroservoelastic ground test. When applicable, this section shall contain the following:

a. Test description.

b. Comparison of the test results with corresponding analysis.

3.2 Component test. When applicable, this section shall contain results of the following tests:

a. Component ground vibration tests on control surfaces, pylons, flaps, horizontal stabilizers, canards, control surface actuators, control surface dampers, racks for supporting stores, weapons bay doors, landing gear doors, and landing gear.

b. Component stiffness tests.

c. Control surface, tab, and actuator rigidity, free play, and wear tests.

d. Balance weight attachment verification tests.

e. Mass property measurements of control surfaces, tabs, and external stores,

f. Flutter damper qualification and wear tests.

4. End of DI-MISC-81387A

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